

BehaviorIQ: Technical Report & Evaluation Analysis

- 1. Dataset Generation & Schema:** Synthetic access log stream simulating 7 attack vectors (Brute Force, Credential Stuffing, Impossible Travel, Lateral Movement, Device Spoofing, Low-and-Slow, Insider Drift) across users, service accounts, and edge devices.
- 2. Model Architecture:** Dual-stage cascade combining Unsupervised Isolation Forest (5,000 trees) for baseline anomaly scoring [0-100] + LightGBM Multi-Class Classifier for threat taxonomy assignment.
- 3. Results & Benchmark Evaluation:** Binary Anomaly F1-Score = 0.961, Precision = 0.972, Recall = 0.951. Top 1% Alert Budget FPR = 0.32%. Model inference latency < 0.8ms per event.
- 4. Cold-Start & Drift Mitigation:** Hierarchical Bayesian Blending reduces cold-start FPR to 0.85% for $N < 20$ events. EMA ($\alpha=0.05$) dynamically adapts baseline profiles without model retraining.
- 5. Explainability (SHAP XAI):** Computes exact SHAP feature attributions and generates natural-language forensic narratives per alert.
- 6. Live Deployments:** Vercel UI (<https://behavior-iq-jade.vercel.app>), Render Backend (<https://behavior-iq.onrender.com>), GitHub (<https://github.com/Yash-Singh607/Behavior-IQ>).